



MATHEMATICS

4 Unit

Year 12

March 2000

Time Allowed: 90 minutes

Instructions:

- There are nine (9) questions
- Attempt all questions
- Questions are not of equal value
- All necessary working should be shown. Marks may be deducted for careless or badly arranged work
- Write on one side of the paper only.

Name:

Question 1

(4) a Factorise the following quadratics into linear complex factors

(i) $x^2 + 2x + 2$

(ii) $4x^2 + 4x + 2$

(8) b If the complex number z is equal to $-2i$, find in cartesian form, and plot on an Argand Diagram, the following complex numbers.

(i) $\bar{z}$

(ii) $z + \bar{z}$

(iii) $z\bar{z}$

(iv) $z + i\bar{z}$

Question 2.

(6) a (i) Find the solution to $\sqrt{16+30i}$

(ii) Solve for complex z if: $z^2 - z - 4 = i(z+7)$

(6) b If w is a cube root of 1, simplify

(i) $1 + 2w + w^2$

(ii) $\frac{1 + 2w + 3w^2}{2 + 3w + w^2}$

(iii) $(2 + 2w^2)^4$

Question 3.

- (6) a) (i) Given de Moivre's Theorem, $(\cos \theta + i \sin \theta)^n = \cos n\theta + i \sin n\theta$ is true for positive integers, show that it is also true for negative integers.
- (ii) If $z = (1 - i\sqrt{3})$, express z^{-4} in the simplest cartesian form.
- (3) b) Sketch the region in the Argand diagram consisting of those points z for which $|\arg z| < \frac{\pi}{3}$, $z + \bar{z} < 4$, and $|z| \geq 2$
- (3) c) Draw the locus of z given $\arg\left(\frac{z-1}{z+1}\right) = \frac{\pi}{4}$

Question 4.

Consider the equation $z^5 = -1$

- (5) a) Solve for complex z and plot the solutions on the Argand Diagram
- (3) b) Express the factors of $z^5 + 1$ over:
- (i) the complex field
 - (ii) the real field
 - (iii) the rational field
- (4) c) Factorise $z^4 - z^3 + z^2 - z + 1$ over the real field and hence or otherwise show that $\cos \frac{\pi}{5} \cos \frac{3\pi}{5} = -\frac{1}{4}$

Question 5.

- (4) a) Let $z_1 = 4 + 8i$, $z_2 = -4 - 8i$
- Show that the locus specified by $|z - z_1| = 3|z - z_2|$ is a circle. Give its centre and radius.
- (3) b) Describe the locus of the complex number w , where $w = \frac{z-4}{z}$ and $|z| = 2$
- (6) c) Sketch $|z - 2i| = 1$
- Find the maximum and minimum value of
- (i) $\arg z$
 - (ii) $|z|$

Question 6.

- (3) a) Using De Moivre's Theorem, express $\sin 3\theta$ and $\cos 3\theta$ in terms of $\sin \theta$ and $\cos \theta$
- (3) b) Hence, express $\tan 3\theta$ in terms of $\tan \theta$
- (3) c) Solve $\tan 3\theta = 1$ for $0 < \theta < \pi$
- (3) d) Hence, explain why $\tan \frac{\pi}{12}$, $\tan \frac{5\pi}{12}$ and $\tan \frac{3\pi}{4}$ are roots of $x^3 - 3x^2 - 3x + 1 = 0$ hence evaluate $\tan \frac{\pi}{12} + \tan \frac{5\pi}{12} + \tan \frac{3\pi}{4}$

Question 7.

- (8) a) (i) Express $\sin^6 \theta$ in the form of $A \frac{\cos}{\sin} 6\theta + B \frac{\cos}{\sin} 4\theta + C \frac{\cos}{\sin} 2\theta + D$
 (ii) Hence, or otherwise, find $\int \sin^6 \theta d\theta$

- (4) b) OABC is a rhombus where $\angle AOC = 60^\circ$.

Given that A represents the complex number $2 + i$, give the complex numbers represented by points B, C.

Question 8.

- (6) a) Two roots of a polynomial with rational coefficients are, $(1 - \sqrt{2})$ and $(2 + i)$. Find the monic polynomial with the lowest degree that satisfies these roots, and express in the form $ax^n + bx^{n-1} + \dots$

- (6) b) Given that $x^3 + px^2 + qx + r = 0$ has roots α, β, γ

(i) Find the equation whose roots are $\alpha^2, \beta^2, \gamma^2$

(ii) Evaluate $\alpha^3 + \beta^3 + \gamma^3$

Question 9.

- (4) a) Solve $x^4 - 6x^3 + 12x^2 - 10x + 3 = 0$ given that the equation has a root of multiplicity 3
 (4) b) The remainder when $x^3 + px + q$ is divided by $(x - 2)(x + 3)$ is $2x + 1$. Find p and q .
 (5) c) w is a cube root of 1. Let α and β be real numbers. Find in its simplest form the cubic equation whose roots are $\alpha + \beta, \alpha w + \beta w^{-1}, \alpha w^2 + \beta w^{-2}$

Question 1

a) i) $x^2 + 2x + 2 = 0$ (solving)
 $x = \frac{-2 \pm \sqrt{4-8}}{2} = \frac{-2 \pm \sqrt{-4}}{2}$
 $= \frac{-2 \pm 2i}{2} \therefore x = -1 \pm i$

$x^2 + 2x + 2 = [x - (-1+i)][x - (-1-i)]$
 $= (x+1-i)(x+1+i)$

OR $x^2 + 2x + 2 = x^2 + 2x + 1 + 1$
 $= (x+1)^2 + 1$
 $= (x+1)^2 - i^2$
 $= (x+1-i)(x+1+i)$

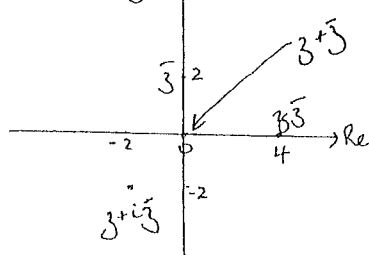
ii) $4x^2 + 4x + 2 = 4x^2 + 4x + 1 + 1$
 $= (2x+1)^2 - i^2$
 $= (2x+1-i)(2x+1+i)$

b) $z = -2i$ $\bar{z} = -2$

i) $\bar{z} = 2i$ ii) $z + \bar{z} = -2i + 2i = 0$

iii) $z\bar{z} = -2i \cdot 2i = 4$

iv) $z + i\bar{z} = -2i - 2$



Question 2

a) i) let $z = \sqrt{16+30i}$
 $z = x+iy$
 $\therefore (x+iy)^2 = 16+30i$

$x^2 - y^2 = 16$
 $and 2ixy = 30i \Rightarrow 2xy = 30$
 $now (x^2+y^2)^2 = (x^2-y^2)^2 + 4x^2y^2$
 $= 16^2 + 30^2$
 $\therefore x^2+y^2 = 34$ ($x^2+y^2 > 0$)

$\therefore x^2+y^2 = 34$ (1)
 $x^2-y^2 = 16$ (2)
 $2x^2 = 50 \Rightarrow x = \pm 5$
 $y = \pm 3 \therefore z = 5+3i$
 $or z = -5-3i$

ii) $z^2 - z - 4 = iz + 7i$
 $z^2 - z(1+i) - (4+7i) = 0$
 $z = \frac{(1+i) \pm \sqrt{(1+i)^2 + 4(4+7i)}}{2}$
 $= \frac{(1+i) \pm \sqrt{2i + 16 + 28i}}{2}$
 $= \frac{(1+i) \pm \sqrt{16+30i}}{2}$
 $= \frac{(1+i) \pm (5+3i)}{2}$

$\therefore z = \frac{6+4i}{2} = 3+2i$ or $\frac{-4-2i}{2} = -2-i$

b) $\omega^3 = 1$ and $1+\omega+\omega^2 = 0$

i) $1+2\omega+\omega^2 = 1+2\omega+(1-\omega)$
 $= \omega$

ii) $\frac{1+2\omega+3\omega^2}{2+3\omega+\omega^2} = \frac{\omega(1+2\omega+3\omega^2)}{\omega(2+3\omega+\omega^2)}$
 $= \frac{\omega(1+2\omega+3\omega^2)}{2\omega+3\omega^2+\omega^3} = \omega$

iii) $2^4(1+\omega)^4 = 16(-\omega)^4 = 16\omega^4$
 $= 16\omega \cdot \omega^3 = 16\omega$

Question 3

a) i) $(\cos \theta + i \sin \theta)^n$
 $= \cos n\theta + i \sin n\theta$

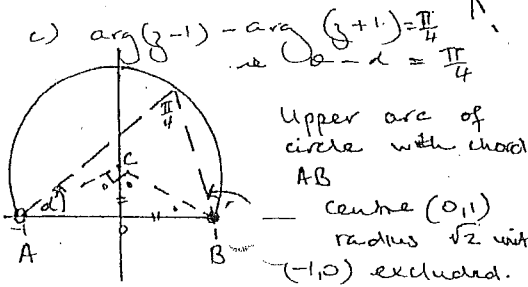
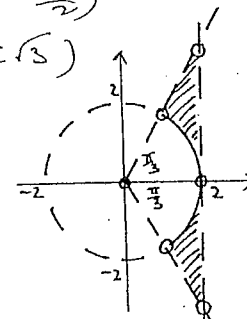
let $n = -m$ where m is positive
 $(\cos \theta + i \sin \theta)^{-m} = \cos -m\theta + i \sin -m\theta$
 $LHS: \frac{1}{(\cos \theta + i \sin \theta)^m} = \frac{1}{\cos m\theta + i \sin m\theta}$

$= \frac{\cos m\theta - i \sin m\theta}{\cos^2 m\theta + \sin^2 m\theta}$
 $= \cos m\theta - i \sin m\theta$

Also: $\cos -m\theta = \cos m\theta$ (even fn)
 $\sin -m\theta = -\sin m\theta$ (odd fn)
 $\therefore RHS = \cos m\theta - i \sin m\theta$
 $= LHS$

ii) $z = 1-i\sqrt{3}$ $|z| = \sqrt{1+3} = 2$
 $arg z = \theta$ where $\tan \theta = -\sqrt{3}$
 $\therefore arg z = -\frac{\pi}{3}$
 $\therefore z^{-4} = [2(\cos -\frac{\pi}{3} + i \sin -\frac{\pi}{3})]^{-4}$
 $= \frac{1}{2^4} (\cos + \frac{4\pi}{3} + i \sin \frac{4\pi}{3})$
 $= \frac{1}{16} (-\frac{1}{2} - i\frac{\sqrt{3}}{2})$
 $= -\frac{1}{32} (1+i\sqrt{3})$

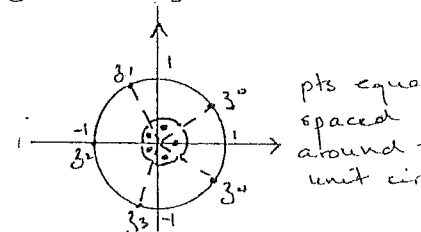
b) $z + \bar{z} < 4$
 $2x < 4$
 $x < 2$



Question 4

$z^5 = -1$
 $let z = r(\cos \theta + i \sin \theta)$ (by De Moivre)
 $z^5 = r^5(\cos 5\theta + i \sin 5\theta)$
 $-1 = 1(\cos \pi + i \sin \pi) \therefore r = 1$
 $\therefore \cos 5\theta = \cos (\pi + 2k\pi)$
 $5\theta = \pi + 2k\pi$
 $\theta = \frac{\pi + 2k\pi}{5}$

$\therefore z_0 = \cos \frac{\pi}{5} + i \sin \frac{\pi}{5}$
 $z_1 = \cos \frac{3\pi}{5} + i \sin \frac{3\pi}{5}$
 $z_2 = \cos \frac{5\pi}{5} + i \sin \frac{5\pi}{5} = -1$
 $z_3 = \cos \frac{7\pi}{5} + i \sin \frac{7\pi}{5} = \bar{z}_1$
 $z_4 = \cos \frac{9\pi}{5} + i \sin \frac{9\pi}{5} = \bar{z}_0$



b) $z^5 + 1 = (z+1)(z-z_0)(z-z_1)(z-z_2)(z-z_3)(z-z_4)$
 $= (z+1)(z-z_0)(z-z_1)(z-z_2)(z-z_3)(z-z_4)$
 $= (z+1)(z^2 - z(z_0+z_4) + z_0z_4)(z^2 - z(z_1+z_3) + z_1z_3)(z-z_2)$
 $= (z+1)(z^2 - 2\cos \frac{\pi}{5}z + 1)(z^2 - 2\cos \frac{3\pi}{5}z + 1)(z-z_2)$
 $= (z+1)(z^4 - z^3 + z^2 - z + 1)$ by long division

c) $z^4 - z^3 + z^2 - z + 1$
 $= (z^2 - 2\cos \frac{\pi}{5}z + 1)(z^2 - 2\cos \frac{3\pi}{5}z + 1)$
 $let z^2 = -1$ i.e. $z = i$
 $(-1)^2 - (-1)i + -1 - i + 1$
 $= (-2\cos \frac{\pi}{5}i)(-2\cos \frac{3\pi}{5}i)$
 $1 + i^2 = -2i + 1 = 4\cos \frac{\pi}{5} \cos \frac{3\pi}{5}$
 $\therefore \cos \frac{\pi}{5} \cos \frac{3\pi}{5} = \frac{1}{4}$

or equate coeffs of z^2 in (c)
 $1 = 1 + 4\cos \frac{\pi}{5} \cos \frac{3\pi}{5} + 1$
 $1 = 2 + 4\cos \frac{\pi}{5} \cos \frac{3\pi}{5}$
 $\cos \frac{\pi}{5} \cos \frac{3\pi}{5} = \frac{1}{4}$

$$k) |z - (4+8i)| = |(x-4) + i(y-8)|$$

$$= \sqrt{(x-4)^2 + (y-8)^2}$$

$$|z - (-4-8i)| = |(x+4) + i(y+8)|$$

$$= \sqrt{(x+4)^2 + (y+8)^2}$$

now:

$$(x-4)^2 + (y-8)^2 = 9[(x+4)^2 + (y+8)^2]$$

$$x^2 - 8x + 16 + y^2 - 16y + 64 = 9x^2 + 72x + 144 + 9y^2 + 144y + 576$$

$$\therefore 8x^2 + 80x + 8y^2 + 160y = 80 - 720$$

$$x^2 + 10x + y^2 + 20y = -80$$

$$(x+5)^2 + (y+10)^2 = -80 + 25 + 100 = 45$$

which represents a circle with centre $(-5, -10)$ rad = $3\sqrt{5}$

b) $w = \frac{z-4}{z}$

$$wz = z - 4$$

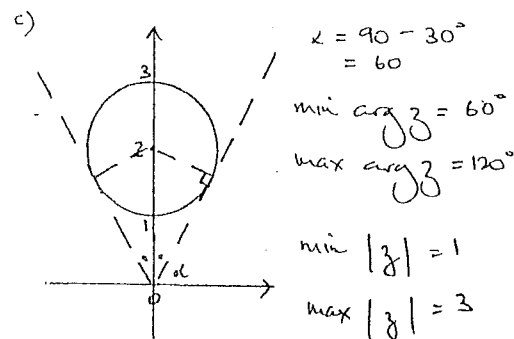
$$4 = z - wz$$

$$= z(1-w)$$

$$|4| = |z||1-w|$$

$$\therefore 2|w-1| = 4 \Rightarrow |w-1| = 2$$

circle, centre $(1, 0)$ rad = $2u$



Question 7

a) $(\cos \theta + i \sin \theta)^3 = \cos 3\theta + i \sin 3\theta$

$$(\cos \theta + i \sin \theta)^3 = \cos^3 \theta + 3 \cos^2 \theta (i \sin \theta) + 3 \cos \theta (i \sin \theta)^2 + (i \sin \theta)^3$$

$$\therefore \cos 3\theta = \cos^3 \theta - 3 \cos \theta \sin^2 \theta$$

$$\sin 3\theta = 3 \cos^2 \theta \sin \theta - \sin^3 \theta$$

b) $\tan 3\theta = \frac{\sin 3\theta}{\cos 3\theta}$

$$= \frac{3 \cos^2 \theta \sin \theta - \sin^3 \theta}{\cos^3 \theta - 3 \cos \theta \sin^2 \theta}$$

$$= \frac{3 \tan \theta - \tan^3 \theta}{1 - 3 \tan^2 \theta}$$

c) $\tan 3\theta = 1$

$$3\theta = \frac{\pi}{4}, \frac{5\pi}{4}, \frac{9\pi}{4}$$

$$\theta = \frac{\pi}{12}, \frac{5\pi}{12}, \frac{9\pi}{12}$$

d) $\tan 3\theta = 1$

$$\therefore \frac{3 \tan \theta - \tan^3 \theta}{1 - 3 \tan^2 \theta} = 1$$

$$3 \tan \theta - \tan^3 \theta = 1 - 3 \tan^2 \theta$$

$$\tan^3 \theta - 3 \tan^2 \theta - 3 \tan \theta + 1 = 0$$

let $\tan \theta = x$

$$x^3 - 3x^2 - 3x + 1 = 0$$

solutions are $x = \tan \theta$

i.e. $x = \tan \frac{\pi}{12}$

$$x = \tan \frac{5\pi}{12}$$

$$x = \tan \frac{3\pi}{4}$$

now sum of roots = 3

$$\therefore \tan \frac{\pi}{12} + \tan \frac{5\pi}{12} + \tan \frac{3\pi}{4} = 3$$

Question 7

a) $z + \frac{1}{z} = 2 \cos \theta$

$$z - \frac{1}{z} = 2i \sin \theta$$

$$z^2 + \frac{1}{z^2} = 2 \cos 2\theta$$

$$\left(z - \frac{1}{z}\right)^2 = (2i \sin \theta)^2 = -4 \sin^2 \theta$$

$$= -4 (i^2) \sin^2 \theta = -4 \sin^2 \theta$$

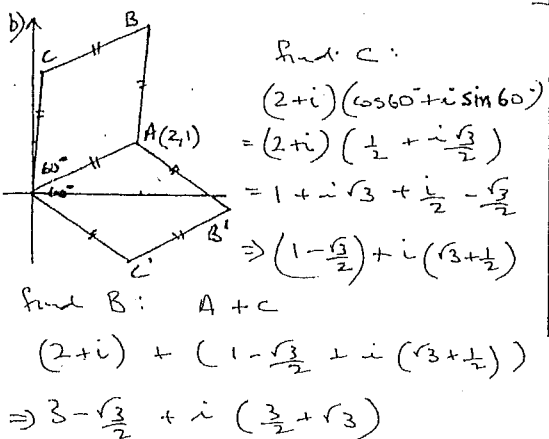
$$\left(z - \frac{1}{z}\right)^6 = z^6 - 6z^5 \cdot \frac{1}{z} + 15z^4 \cdot \frac{1}{z^2} - 20z^3 \cdot \frac{1}{z^3} + 15z^2 \cdot \frac{1}{z^4} - 6z \cdot \frac{1}{z^5} + \frac{1}{z^6}$$

$$= z^6 - 6z^4 + 15z^2 - 20 + 15z^{-2} - 6z^{-4} + z^{-6}$$

$$= z^6 + \frac{1}{z^6} - 6\left(z^4 + \frac{1}{z^4}\right) + 15\left(z^2 + \frac{1}{z^2}\right) - 20$$

$$\sin^6 \theta = \frac{-1}{64} (2 \cos 6\theta - 12 \cos 4\theta + 30 \cos 2\theta - 20)$$

$$\int \sin^6 \theta d\theta = \frac{-1}{64} \left(\frac{2 \sin 6\theta}{6} - 3 \sin 4\theta + 15 \sin 2\theta - 20\theta \right)$$



Question 8

a) roots $(1-\sqrt{2}), (2+i) \therefore$ also $(2-i)$. coeffs are real, also $(1+\sqrt{2})$

$$\Rightarrow (x-1+\sqrt{2})(x-1-\sqrt{2})(x-2-i)(x-2+i)$$

$$= [(x-1)^2 - 2][(x-2)^2 - i^2]$$

$$= (x^2 - 2x + 1 - 2)(x^2 - 4x + 4 + 1)$$

$$= (x^2 - 2x - 1)(x^2 - 4x + 5)$$

$$= x^4 - 4x^3 + 5x^2 - 2x^3 + 8x^2 - 10x - x^2 + 4x - 5$$

$$= x^4 - 6x^3 + 12x^2 - 6x - 5$$

poly. is $x^4 - 6x^3 + 12x^2 - 6x - 5$

b) $y = x^2 \Rightarrow x = y^{\frac{1}{2}}$ etc.

$$(y^{\frac{1}{2}})^3 + p(y^{\frac{1}{2}})^2 + q(y^{\frac{1}{2}}) + r = 0$$

$$y^{\frac{3}{2}} + py + qy^{\frac{1}{2}} + r = 0$$

$$y^{\frac{1}{2}}(y + q) = -py - r$$

$$y(y + q)^2 = (-py - r)^2$$

$$y(y^2 + 2qy + q^2) = p^2y^2 + 2pry + r^2$$

$$y^3 + y^2(2q - p^2) + y(q^2 - 2pr) - r^2 = 0$$

$$\therefore x^3 + x^2(2q - p^2) + x(q^2 - 2pr) - r^2 = 0$$

ii) $x^3 + px^2 + qx + r = 0$

$$p^3 + p^2 + q^2 + r = 0$$

$$r^3 + pr^2 + qr + r = 0$$

$$x^3 + p^3 + r^3 + p(x^2 + p^2 + r^2) + q(x + p + r) + 3r = 0$$

$$x^3 + p^3 + r^3 = -p(p^2 + 2q) - q(-p) - 3r$$

$$= -p^3 + 3pq - 3r$$

c) (i) 4th quad: $(2+i)(\cos 60^\circ + i \sin 60^\circ)$

$$= (2+i)\left(\frac{1}{2} + i\frac{\sqrt{3}}{2}\right) = \left(1 + \frac{\sqrt{3}}{2}\right) + i\left(\frac{3}{2} + \sqrt{3}\right)$$

B': $(2+i) + \left(1 + \frac{\sqrt{3}}{2} + i\left(\frac{3}{2} + \sqrt{3}\right)\right)$

$$= \left(3 + \frac{\sqrt{3}}{2}\right) + i\left(\frac{3}{2} + \sqrt{3}\right)$$

Question 9.

$$a) P(x) = x^4 - 6x^3 + 12x^2 - 10x + 3$$

$$P'(x) = 4x^3 - 18x^2 + 24x - 10$$

$$P''(x) = 12x^2 - 36x + 24$$

$$\text{mult. 3} \Rightarrow P''(x) = 0 \text{ gives the root}$$

$$12(x^2 - 3x + 2) = 0$$

$$12(x-2)(x-1) = 0$$

$$x = 2 \text{ or } 1$$

$$P(2) = 16 - 48 + 24 - 20 + 3 \neq 0$$

$$P(1) = 1 - 6 + 12 - 10 + 3 = 0$$

$$\therefore x = 1 \text{ is the triple root}$$

$$P(x) = (x-1)^3 Q(x) = (x-1)^3 (x+b)$$

$$\text{equating coeff of } x^4$$

$$1 = a$$

$$\text{equating constants}$$

$$3 = -b \therefore b = -3$$

$$P(x) = (x-1)^3 (x-3) = 0$$

$$x = 1, 1, 1, 3$$

$$b) x^3 + px + q = (x-2)(x+3)Q(x) + 2x+1$$

$$x=2$$

$$8 + 2p + q = 4 + 1$$

$$x=3$$

$$-27 - 3p + q = -6 + 1$$

$$\therefore 2p + q = -3 \quad (1)$$

$$-3p + q = 22 \quad (2)$$

$$(1)-(2) \quad 5p = -25$$

$$p = -5$$

$$q = 7$$

$$\therefore \text{eqn: } x^3 - 0x^2 - 3\alpha\beta x - (\alpha^3 + \beta^3) = 0 \text{ ie } x^3 - 3\alpha\beta x - (\alpha^3 + \beta^3) = 0$$

$$c) \omega^3 = 1$$

$$1 + \omega + \omega^2 = 0$$

$$\Sigma \alpha$$

$$= \alpha + \beta + \alpha\omega + \frac{\beta}{\omega} + \alpha\omega^2 + \frac{\beta}{\omega^2}$$

$$= \alpha(1 + \omega + \omega^2) + \beta(1 + \frac{1}{\omega} + \frac{1}{\omega^2})$$

$$= 0 + \beta \left(\frac{\omega^2 + \omega + 1}{\omega^2} \right) = 0$$

$$\Sigma \alpha\beta$$

$$(\alpha + \beta)(\alpha\omega + \frac{\beta}{\omega}) + (\alpha + \beta)(\alpha\omega^2 + \frac{\beta}{\omega^2})$$

$$+ (\alpha\omega + \frac{\beta}{\omega})(\alpha\omega^2 + \frac{\beta}{\omega^2})$$

$$= \alpha^2\omega + \frac{\alpha\beta}{\omega} + \alpha\beta\omega + \frac{\beta^2}{\omega}$$

$$+ \alpha^2\omega^2 + \frac{\alpha\beta}{\omega^2} + \alpha\beta\omega^2 + \frac{\beta^2}{\omega^2}$$

$$+ \alpha^2\omega^3 + \frac{\alpha\beta}{\omega} + \alpha\beta\omega + \frac{\beta^2}{\omega^3}$$

$$= \alpha^2(\omega + \omega^2 + \omega^3) + \alpha\beta(\frac{1}{\omega} + \frac{1}{\omega^2} + \frac{1}{\omega^3})$$

$$+ \alpha\beta(\omega + \omega^2 + \omega) + \beta^2(\frac{1}{\omega} + \frac{1}{\omega^2} + \frac{1}{\omega^3})$$

$$= \alpha^2(0) + \alpha\beta \left(\frac{2\omega + 1}{\omega^2} \right) + \beta^2 \left(\frac{2\omega + 1}{\omega^2} \right)$$

$$+ \beta^2 \left(\frac{\omega^2 + \omega + 1}{\omega^3} \right)$$

$$= \alpha\beta(\omega^2 - 1 + 2\omega + \omega^2)$$

$$= \alpha\beta(2\omega^2 + 2\omega - 1)$$

$$= \alpha\beta(2(-1) - 1) = -3\alpha\beta$$

$$\Sigma \alpha\beta\gamma$$

$$(\alpha + \beta)(\alpha\omega + \frac{\beta}{\omega})(\alpha\omega^2 + \frac{\beta}{\omega^2})$$

$$= (\alpha + \beta)(\alpha^2\omega^3 + \frac{\alpha\beta}{\omega} + \alpha\beta\omega + \frac{\beta^2}{\omega^3})$$

$$= \alpha^3\omega^3 + \frac{\alpha^2\beta}{\omega} + \alpha^2\beta\omega + \frac{\alpha\beta^2}{\omega^3}$$

$$+ \alpha^2\beta\omega^3 + \frac{\alpha\beta^2}{\omega} + \alpha\beta^2\omega + \frac{\beta^3}{\omega^3}$$

$$= \alpha^3 + \alpha^2\beta \left(\frac{1}{\omega} + \omega + \frac{1}{\omega^3} \right)$$

$$+ \alpha\beta^2 \left(\frac{1}{\omega} + \omega + \frac{1}{\omega^3} \right) + \beta^3$$

$$= \alpha^3 + \beta^3$$

$$\text{NB: } \frac{\omega^2 + \omega + 1}{\omega^3}$$

$$= \frac{\omega^2 + \omega + 1}{\omega^3}$$

$$= 0$$